Sl.No	Title and Year	Key Findings	Datasets	Methodology	Gaps
1	TruthfulRAG: Resolving Factual-level Conflicts in Retrieval-Augmented Generation with Knowledge Graphs	Knowledge-graph reasoning paths and entropy-based filtering improve factual accuracy and context precision when retrieved evidence conflicts with the LLM's internal knowledge.	FaithEval, MuSiQue, RealtimeQA, SQuAD; MuSiQue-golden and SQuAD-golden for non-conflicting-context evaluation.	Construct a knowledge graph from retrieved text; retrieve query-relevant graph paths using entity and relation matching; use entropy changes to identify potentially conflicting evidence; generate answers using selected reasoning paths.	Fixed entropy thresholds may not generalize across models and datasets. Graph construction and path retrieval add computational overhead, while errors in extracted triples can affect results. Entropy changes alone do not conclusively establish factual conflict.
2	Toward Robust GraphRAG: Mitigating Retrieval Drift and Hallucination from Imperfect Knowledge Graphs. (2026)	The proposed CS-RAG framework improves robustness across different knowledge-graph builders and under controlled graph-quality degradation.It reduces retrieval drift caused by incorrect triples and retrieval hallucination caused by insufficient structural evidence.	MuSiQue, 2WikiMultihopQA, HotpotQA. The experiments use 1,000 validation instances per benchmark.	CS-RAG converts questions into ordered atomic constraints with intermediate variables and relation variants. It retrieves and reranks graph candidates using entity anchors and relation matching, checks evidence sufficiency using an effective-candidate distribution score, and uses textual recovery when graph evidence is insufficient. The selected evidence is passed to an LLM to generate the final answer.	Existing GraphRAG approaches often assume that knowledge graphs are reliable. Noisy triples can cause retrieval drift, while missing edges and dropped qualifiers can lead to unsupported reasoning chains. CS-RAG also remains vulnerable to incorrect constraint planning and cases where the required bridge evidence is absent from both the graph and recovered text.
3	SELF-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection.(2024)	SELF-RAG outperformed several baseline LLMs and conventional RAG systems on multiple question-answering, reasoning, fact-verification, and long-form generation tasks. It improved factuality and citation precision and allowed retrieval behavior to be adjusted at inference time.	PubHealth, ARC-Challenge, PopQA, TriviaQA-unfiltered, biography generation dataset, and ALCE-ASQA.	Trains an LLM to use special reflection tokens to decide when to retrieve evidence, assess retrieved passage relevance, evaluate whether generated claims are supported, and score response usefulness. Candidate outputs are ranked using critique scores during decoding.	The paper acknowledges that outputs can still contain claims that are not fully supported by citations. Self-assessment may be imperfect, and performance depends on retrieval quality and the model's ability to judge relevance and factual support.
4	Corrective Retrieval Augmented Generation(2024)	CRAG improves standard RAG and Self-RAG by evaluating retrieval quality, refining relevant evidence, and supplementing poor retrieval results with web search. It improves performance across short-form and long-form generation tasks.	PopQA, Biography, PubHealth, ARC-Challenge	A lightweight T5-large retrieval evaluator assigns confidence scores and triggers three actions: Correct, Incorrect, or Ambiguous. Relevant documents are decomposed into smaller knowledge segments and filtered. When retrieval is unreliable, web search provides additional evidence. A generator then produces the final response using the refined evidence.	The method requires fine-tuning an external retrieval evaluator, and its confidence thresholds are empirically selected. Retrieval evaluation errors and unreliable external web sources may still affect factual reliability.
5	LettuceDetect: A Hallucination Detection Framework for RAG Applications(2025)	LettuceDetect-large achieved an overall example-level F1 score of 79.22% and a span-level F1 score of 58.93% on RAGTruth. It outperformed the encoder-based model Luna (65.4% example-level F1) and most prompt-based methods. The models process approximately 30–60 examples per second on a single GPU.	RAGTruth benchmark dataset, containing approximately 18,000 span-annotated examples across question answering, data-to-text generation, and news summarization. Its source datasets include MS MARCO, Yelp Open Dataset, and CNN/Daily Mail.	Uses ModernBERT-base and ModernBERT-large as encoder backbones for token classification. The model processes context, question, and answer together, classifies answer tokens as supported or hallucinated, and combines consecutive hallucinated tokens into spans. Training uses AdamW, a learning rate of 1×10^{-5}, weight decay of 0.01, batch size 8, and six epochs.	The evaluation primarily uses the RAGTruth benchmark. The authors identify future work involving additional datasets, more languages, and enhanced architectures. The approach uses a fixed 0.5 confidence threshold for span aggregation and binary hallucination labels rather than distinguishing hallucination categories.
6	RT4CHART: Retromorphic Testing for Hallucination Detection in RAG Pipelines (2026)	Achieved answer-level F1 of 0.776 on RAGTruth++ and 0.845 on RAGTruth-Enhance. Achieved span-level F1 of 40.4% and 47.5%, respectively. Claim-based local verification contributed most of the improvement. The re-annotation identified 1.68× more hallucination cases than the original annotations.	RAGTruth++ (408 samples) and RAGTruth-Enhance (2,675 samples), covering question answering, summarization, and data-to-text tasks.	Decomposes answers into atomic claims; performs local verification against overlapping context chunks; applies global verification against the full context; labels claims as Entailed, Contradicted, or Baseless; maps suspicious answer spans to supporting or contradicting context evidence; aggregates claim-level labels into an answer-level verdict	Despite improved detection, span-level localization and evidence grounding remain imperfect. The approach depends on LLM-based claim decomposition and verification, can struggle with changes in certainty and internally inconsistent context, and evaluates faithfulness strictly against retrieved context. Generalization to other domains and languages remains unverified.
7	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks(2020)	RAG combines parametric memory (BART) with non-parametric memory (a Wikipedia document index). It achieved state-of-the-art results on four open-domain QA benchmarks at the time and generated more factual, specific, and diverse responses than the BART-only baseline. The external knowledge index can be updated without retraining the entire model.	Natural Questions (NQ), TriviaQA (TQA), WebQuestions (WQ), CuratedTrec (CT), MS MARCO, SearchQA/Jeopardy question generation, and FEVER.	Uses a Dense Passage Retriever (DPR) to retrieve relevant Wikipedia passages and a BART-large generator to generate answers. The retrieved documents are treated as latent variables and marginalized during generation. Two variants are proposed: RAG-Sequence, which uses the same document for the whole output, and RAG-Token, which can use different documents for different tokens. The retriever and generator are fine-tuned jointly.	Although RAG improves factuality, it does not eliminate hallucinations. Its performance depends on the relevance and reliability of retrieved documents, and the model may generate unsupported claims when retrieval fails or the knowledge source contains incorrect or biased information. The paper does not directly provide a comprehensive answer-span hallucination localization and evidence-grounding mechanism. Retrieval collapse is also observed in some generation tasks.
8	Retrieval-Augmented Generation: A Comprehensive Survey of Architectures, Enhancements, and Robustness Frontiers(2025)	Retriever-based models excel in multi-hop QA; Generator models excel in short-form QA; Hybrid models achieve highest factual consistency; Noise paradox (random docs can improve reasoning); Inherent trade-offs between precision, efficiency, and faithfulness.	Short-form: NQ, TriviaQA, PopQA, ARC-Challenge, WebQA, SciQ. Multi-hop: HotpotQA, 2Wiki, MuSiQue, StrategyQA. Robustness/ Specialty: FEVER, ASQA, MS MARCO, KILT-WoW, Biography, RGB, RAG-Bench, RAGTruth, MIRAGE, FeB4RAG, RetrievalQA.	Taxonomy creation of 4 RAG paradigms; component analysis across 5 enhancement dimensions; comparative benchmarking normalizing for backbone models; synthesis of automated evaluation frameworks.	Need for dynamic retrieval adaptivity; defenses against adversarial corpus poisoning; structured compositionality for long-range multi-turn reasoning; meta-retrievers for cross-domain/temporal drift; privacy-preserving and explainable retrieval systems.

9	Language Models are Few-Shot Learners(2020)	Scaling up language models significantly improves task-agnostic, few-shot performance. GPT-3 performs competitively or exceeds fine-tuned state-of-the-art models on several tasks using only text context. Humans struggle to distinguish GPT-3 generated news articles from human-written ones.	PTB, LAMBADA, HellaSwag, StoryCloze, Natural Questions, WebQuestions, TriviaQA, WMT'14/16 (Fr/De/Ro), Winograd, Winogrande,PIQA, ARC, OpenBookQA, CoQA, DROP, QuAC, SQuADv2, RACE, SuperGLUE, ANLI, RTE.	Trained a 175B parameter autoregressive language model (GPT-3). Evaluated on over two dozen NLP benchmarks in zero-shot, one-shot, and few-shot settings without any gradient updates. Analyzed performance scaling across 8 model sizes (125M to 175B) and conducted N-gram overlap analysis for test set contamination.	Lacks bidirectional context capabilities; loses coherence in very long generation; lacks physical world grounding; poor pre-training sample efficiency; highly expensive/inconvenient inference requiring model distillation; inherits severe societal biases from training data.
10	Mitigating Hallucination in Large Language Models: A Capability-Oriented Survey on RAG, Reasoning, and Agentic Systems(2025)	RAG mitigates knowledge-based hallucinations via broad and precise retrieval pipelines. Reasoning enhancements (CoT, Tool-augmented, Symbolic) mitigate logic-based hallucinations by improving logical deduction and constraints. Agentic systems integrate retrieval, multi-step reasoning, and iterative feedback loops to effectively mitigate composite hallucinations in complex, multi-stage tasks.	TruthfulQA, FreshQA, MedHallu, HalluLens, RAGTruth (Knowledge-based); BIG-bench, PrOntoQA, ProofWriter, ToolBench, API-Bank, LogicBench, ReClor, LogiQA (Logic-based); AgentBench, L-MARS, InfoDeepSeek, R-Judge (Agentic/Composite).	Conducted a systematic, capability-oriented survey of inference-time hallucination mitigation techniques. Categorized hallucinations into "knowledge-based" and "logic-based." Analyzed mitigation methods across three paradigms: RAG (for knowledge acquisition), Reasoning (for reasoning reliability), and Agentic Systems (for capability coordination).	RAG struggles with evaluating and integrating conflicting or noisy external knowledge. LLM reasoning lacks explicit structured world models and intrinsic verification. Agentic systems rely too heavily on predefined workflows and lack dynamic coordination for decisions. There is a high computational cost trade-off, and a severe lack of fine-grained, process-level capability evaluation benchmarks (most only evaluate final outputs).
11	Title: A Comprehensive Survey of Knowledge Graph-Augmented Generation (Graph-RAG) for Trustworthy Large Language Models. Author: Jianwei Ren	Traditional Retrieval-Augmented Generation (RAG) attempts to address this problem by retrieving relevant documents before generating an answer. However, ordinary RAG systems still have difficulties with complex reasoning and retrieving precise facts from long documents	The paper reviews existing research involving knowledge graphs, graph-based retrieval, and Graph-RAG methods. Because this is a survey, we should not claim that the authors trained and evaluated one new Graph-RAG model on a single dataset.	The paper surveys Graph-RAG techniques, organizing them into graph indexing, graph-guided retrieval, and graph-enhanced generation. It compares approaches and discusses their contributions to trustworthy language-model responses.	Challenges include scalable graph construction, incomplete or inaccurate knowledge graphs, complex reasoning, evaluation of reasoning paths, and limited support for causal relationships and reliable answers in domain-specific applications.
12	Hallucination Mitigation for Retrieval-Augmented Large Language Models: A Review — 2025.	Retrieval-Augmented Generation reduces some hallucinations but cannot eliminate them. Errors can originate during retrieval or generation, making evidence quality, factual consistency, hallucination detection, and answer correction important for reliable responses.	This is a review paper, not an original experimental study. It discusses existing research on hallucination mitigation, so there is no single dataset used throughout the paper.	The authors review hallucination causes and mitigation techniques, covering retrieval improvements, generation strategies, prompting, hallucination detection, and correction methods for Retrieval-Augmented Generation systems.	Persistent hallucinations, conflicting evidence, long-context limitations, and unreliable detection remain challenges. Existing techniques lack a universal solution that consistently identifies and corrects unsupported claims across different domains
13	Efficient and Robust Approximate Nearest Neighbor Search Using Hierarchical Navigable Small World Graphs — 2018.	Hierarchical Navigable Small World graphs enable efficient approximate nearest-neighbor search. Their hierarchical structure supports fast navigation through high-dimensional vector spaces while maintaining a useful balance between retrieval speed and search accuracy.	The study evaluates vector-search performance using synthetic and real-world high-dimensional datasets. Experiments compare retrieval efficiency and accuracy against existing approximate nearest-neighbor methods on different data collections.	The method constructs a multilayer proximity graph. Searches begin at upper layers and progressively move toward lower layers, locating approximate nearest neighbors efficiently while controlling search accuracy and computational cost.	Approximate search may miss relevant neighbors. Vector similarity does not guarantee factual correctness or logical relationships, and efficient retrieval alone cannot ensure that a language model generates accurate, evidence-supported answers.
14	Active Retrieval Augmented Generation — 2023.	The proposed FLARE method retrieves information dynamically during generation. Instead of retrieving documents only at the beginning, it identifies uncertain upcoming content and retrieves additional evidence to improve long-form answer generation.	The paper evaluates FLARE across four long-form, knowledge-intensive tasks. The exact dataset names are not specified in the information currently available, so they require verification from the original paper.	FLARE monitors generated text for low-confidence tokens and predicts upcoming sentences. When additional evidence is needed, it formulates retrieval queries, retrieves relevant documents, and continues generation using the supporting information.	Low model confidence does not always indicate factual errors. Dynamic retrieval can increase computational costs and may retrieve conflicting evidence, while confident but unsupported statements can still pass through without sufficient verification.
15	Larger Language Models Do In-Context Learning Differently — 2023.	Larger language models can learn arbitrary input-output label mappings from examples more effectively. Instruction tuning also influences in-context learning, but its effects vary across tasks and do not consistently improve every capability.	The study evaluates seven NLP tasks, including sentiment analysis, subjectivity classification, question classification, paraphrase detection, textual entailment, and hate-speech classification, using multiple language models and task-specific examples.	The researchers compare language models of different sizes using in-context examples, arbitrary label mappings, and instruction-tuned models. They investigate how model scale and training approaches influence learning from prompts.	Following examples does not guarantee factual accuracy or evidence verification. Performance depends on task design and prompts, leaving questions about how models reliably use domain-specific evidence and avoid unsupported answers.
16	Lost in the Middle: How Language Models Use Long Contexts — 2023.	Language models often perform better when relevant information appears at the beginning or end of a context. Evidence positioned in the middle may receive less attention, even when models support long input sequences.	The study uses multi-document question answering and synthetic key-value retrieval tasks. It evaluates language models including GPT-3.5-Turbo, Claude, MPT-30B-Instruct, and LongChat-13B under different context lengths and evidence positions.	The researchers systematically vary the position of relevant information within long contexts and measure model performance. They compare evidence retrieval and question-answering accuracy to identify positional effects on information use.	Longer context windows do not guarantee better evidence utilization. Models may overlook relevant information or mishandle conflicting sources, highlighting the need for improved evidence positioning, retrieval strategies, and claim-level verification.
17	Hallucination Mitigation for Retrieval-Augmented Large Language Models: A Review — 2025. This PDF duplicates Paper 12 and therefore does not represent an independent research contribution.	The review identifies retrieval errors and generation errors as major causes of hallucinations. It emphasizes improving evidence quality, factual consistency, detection, and correction to make Retrieval-Augmented Generation more reliable.	No single original dataset is introduced because this is a review paper. It examines existing studies and their evaluation approaches concerning hallucination mitigation in Retrieval-Augmented Generation systems.	The authors organize and analyze existing hallucination mitigation research, including retrieval optimization, generation improvements, prompting techniques, hallucination detection, and correction strategies for improving factual reliability.	The review highlights unresolved challenges involving conflicting evidence, long contexts, persistent hallucinations, and unreliable correction. Existing methods do not provide a universal guarantee that every generated claim is supported by evidence.

18	A Comprehensive Survey of Knowledge Graph-Augmented Generation (Graph-RAG) for Trustworthy Large Language Models — 2026.	Graph-RAG uses structured knowledge graphs to improve factual grounding, multi-hop reasoning, and answer explainability. Explicit entities, relationships, and reasoning paths can help language models connect information distributed across multiple documents.	The paper is a survey rather than an original dataset-based experiment. It reviews existing Graph-RAG research and discusses public knowledge graphs and evaluation challenges, rather than reporting results from one newly introduced dataset.	The survey categorizes Graph-RAG into graph indexing, graph-guided retrieval, and graph-enhanced generation. It reviews knowledge extraction, graph completion, representation learning, hybrid retrieval, and emerging approaches to causal and generative graph reasoning.	Challenges include scalability, graph construction errors, retrieval latency, incomplete knowledge, and evaluation difficulties. Further research should improve causal reasoning, dynamic knowledge updates, evidence-path verification, and integration of graph structure with natural-language context.
19	Knowledge Graph-Extended Retrieval Augmented Generation for Question Answering — 2025. Authors: Jasper Linders and Jakub M. Tomczak. Published online on 19 November	The proposed KG-RAG system improves multi-hop question answering by decomposing complex questions into simpler sub-questions. Experiments on MetaQA show improved multi-hop accuracy, although performance may slightly decrease on single-hop questions.	The primary evaluation dataset is MetaQA, which contains knowledge-graph question-answering examples involving different reasoning depths. The paper examines one-hop, two-hop, and three-hop questions and evaluates answers using the Hit@1 accuracy metric	KG-RAG combines knowledge-graph retrieval, question decomposition, in-context learning, and Chain-of-Thought prompting. It retrieves candidate triples, answers sub-questions sequentially, and synthesizes the results into a final answer with an explicit reasoning chain.	The method is evaluated primarily on knowledge-graph question answering, limiting conclusions about broader applications. Its performance depends on entity identification, graph completeness, retrieval quality, and question decomposition; further evaluation across domains is needed.
21	Constructing Datasets for Multi-hop Reading Comprehension Across Documents (2018)	Multi-hop reasoning over documents requires connecting facts across multiple passages; standard single-document reading comprehension models fail when evidence is fragmented across multiple sources.	MedHop, WikiHop (derived from PubMed abstracts and Wikipedia).	Constructed multi-hop question-answering pairs requiring relational reasoning chains across disjoint documents, establishing graph-like textual reasoning paths.	Focused purely on dataset construction and evaluation rather than providing a dynamic retrieval or real-time verification and repair framework.
22	Active Retrieval Augmented Generation (2023)	Continuously retrieving context at fixed intervals causes unnecessary overhead and noise; dynamically predicting when and what to retrieve during generation improves output coherence and factual accuracy.	2WikiMultihopQA, HotpotQA, StrategyQA, PopQA.	Introduced FLARE (Forward-Looking Active Retrieval Augmented Generation), which anticipates future sentences, generates tentative tokens, and uses them as queries to retrieve relevant documents if confidence is low.	Relies strictly on textual vector retrieval without structured graph verification, leaving it vulnerable to hallucinations when retrieved text passages contain conflicting information.
23	Lost in the Middle: How Language Models Use Long Contexts (2023)	LLMs perform best when relevant information is at the beginning or end of the context window; performance drops significantly when key evidence is placed in the middle of long contexts.	Multi-document QA benchmarks (NQ-open, TriviaQA), key-value retrieval tasks.	Evaluated language model performance on question answering and key-value retrieval by systematically varying the position of relevant information within input contexts of varying lengths.	Identifies positional attention bias in transformers but does not propose an active structural graph retrieval or post-generation claim repair mechanism to fix the position sensitivity.
24	FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation (2023)	Atomic claim breakdown enables precise, fine-grained factual evaluation of long-form LLM outputs, exposing hallucinations that sentence-level or token-level metrics miss.	Biography generation dataset (people profiles evaluated against Wikipedia).	Decomposes long-form text outputs into atomic facts, then independently verifies each atomic claim against a knowledge source using automated retrieval and LLM/NLI verifiers.	Designed exclusively for offline evaluation and scoring rather than real-time claim adjudication, localized splicing, or automated response repair during RAG generation.
25	SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models (2023)	Black-box LLMs can detect their own hallucinations without external database retrieval by measuring consistency across multiple stochastically sampled responses.	WikiBio dataset, GPT-3 generated biographies.	Samples multiple stochastic generations for a prompt and measures semantic consistency across samples using NLI, BERTScore, and LLM-prompting metrics to highlight ungrounded spans.	Computationally expensive due to multiple sampling passes; lacks external factual grounding against Knowledge Graphs or authoritative text corpora.
26	A Retrieval-Augmented Generation Framework Based on a Knowledge Graph (2025)	Integrating structured Knowledge Graph entities into RAG pipelines significantly reduces domain-specific hallucinations and improves answer precision over pure vector retrieval.	Domain-specific biomedical and technical QA benchmarks.	Combines vector similarity search over text with subgraph retrieval around recognized domain entities, fusing graph triples directly into the generator context.	Fuses graph and text blindly into a single prompt without checking for internal contradictions between textual evidence and graph paths.
27	A Taxonomy of Knowledge Bases for Retrieval-Augmented Methods (2025)	Different knowledge base structures (unstructured text, relational tables, knowledge graphs) exhibit distinct trade-offs regarding retrieval speed, reasoning depth, and update latency in RAG systems.	Survey across multiple industry and academic RAG evaluation suites.	Systematically categorized and benchmarked diverse knowledge sources according to structural granularity, retrieval indexing paradigms, and suitability for multi-hop QA.	Provides a high-level taxonomy and comparative synthesis without implementing a unified dual-channel adjudication or localized claim splicing engine.
28	Verifiable Legal Contract Question Answering Using Natural Language Processing and Knowledge Graph Cross-Checking (2025)	Cross-checking unstructured legal contract clauses against structured Knowledge Graph rules provides verifiable, auditable legal question-answering with high compliance accuracy.	Domain-specific legal contract clause and compliance query datasets.	Extracts contract entities and obligations into a formal Knowledge Graph, executing natural language queries against both clause text and graph constraints with explicit cross-checking logic.	Tailored strictly to deterministic legal contract rules and clause verification; lacks a generic 9-state adjudication matrix or budget-bounded claim-splicing repair for open domain multi-hop QA.